Research Statement

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Overview: Measuring Representation in Collective Decisions

Many collective decisions are now made with the help of algorithms: a voting rule decides which projects a city funds from its participatory budget, electoral systems turn votes into parliaments, and large language models are being tested for summarizing public consultations and drafting statements that a whole group can agree on. In each case, the question is whether the outcome represents the people it affects. Answering it with evidence requires a way to compare decision problems (are synthetic test cases similar to real ones?) and real data to test against. Computational social choice, the field that studies voting rules with the tools of computer science, long relied on proofs and on experiments with synthetic votes, often drawn uniformly at random, with no general method to check how close these were to real elections.

My work provides both. It falls into three lines, which also structure this statement:

- *A geometry of collective decisions.* I introduced *maps of elections*, which place elections in a common space, using a distance that I proposed, so that similar elections lie close together and one can see which models of synthetic votes resemble real data [C3, J4].
- *Real data and real deployments.* I initiated Pabulib, the largest open repository of participatory budgeting data. It holds over 2,000 elections from 8 countries and has been cited in more than 100 papers. I am also a member of the team that convinced Polish cities to adopt the Method of Equal Shares, a proportional rule for participatory budgets.
- *Toward language.* I now study collective decisions in which text and language models play a role, including simulated deliberation among LLM agents [C28] and the voting record of the UN General Assembly.

As a PhD student, I was the principal investigator of a National Science Centre grant (NCN Preludium) on the *election isomorphism problem*, from which maps of elections grew. My thesis, *Map of Elections* (advisor: Piotr Faliszewski), received the Polish Prime Minister's Award (2025), and I received a START fellowship of the Foundation for Polish Science (2023) and the AAMAS 2023 Best Student Paper Award. I was then a CNRS postdoc at LAMSADE (Paris Dauphine-PSL) with Jérôme Lang (2024–2026), working at the intersection of social choice and machine learning, and I am now a postdoc at the University of Geneva.

At EPFL, I will extend these tools to collective decisions in which opinions are expressed in text and aggregated by AI, such as summaries of public consultations, consensus statements, and AI agents that vote on people's behalf. For example, given the comments that residents submitted in a consultation and an AI-written summary of them, how can we measure which groups the summary leaves out?

Past and Current Research

A geometry of collective decisions. How similar are two elections? A useful distance between elections must ignore what the candidates are called and in which order the voters are listed, since two elections can have the same structure with entirely different candidates (presidential candidates vs. types of sushi). We showed that the most natural such distances, and the related problem of finding one election inside another, are computationally hard [C2, J3, C5, J2]. I proposed instead the *positionwise distance*, which compares two elections by how often each candidate is ranked in each position, after optimally matching the candidates of the two elections. It can be computed efficiently, and it made it possible to draw the *map of elections* [C3]: compute the distances between all pairs in a collection of elections and draw them in the plane, so that similar elections lie close together. We then added landmark elections that make its regions interpretable [C4]; the complete framework appeared in *Artificial Intelligence* [J4].

Maps also help design experiments, because they show which parts of the space an experiment covers, which it misses, and where real elections lie relative to synthetic ones. The same approach works beyond ordinal elections. We built maps of approval elections [C7], of elections of different sizes [C19], of stable roommates and stable matching instances [C9, J1] (Best Student Paper Award), of tournaments [C22], and of fair division instances [C25], and used the geometry for preference learning [C8]. To help others use this methodology, I co-authored a guide to numerical experiments on elections [C15] and released two Python libraries: `mapof`, for drawing maps, and `prefsampling`, for sampling synthetic preferences.

I also study the structure within a single election. Polarization is usually measured in elections where each voter names one party. We introduced indices of agreement, diversity, and polarization for ranked ballots, built on *k-Kemeny*, a generalization of Kemeny's rule that seeks k representative rankings instead of one consensus, and extended them to approval ballots [C10, C27]. We then used them to quantify how diverse structured preference domains are [C24, C26]. In another line of work, we do not design a rule but look for hidden structure in a preference dataset, such as consistent subelections [C13] or the most conflicting pair of candidates [C18]. Closer to political science, we designed voting rules that let allied candidates avoid the spoiler effect [C14] and a nonparametric test for gerrymandering [C16].

Real data and real deployments. In participatory budgeting (PB), residents vote on how to spend part of a city's budget. I initiated Pabulib in 2020, and I built and maintain it [C11]. It is now the largest open library of PB data. Our later work on PB includes learning realistic models of approval voting [C20], measuring how well individual projects performed under a given rule [C17], and a game-theoretic analysis of how proposers set project costs under the Method of Equal Shares [C23].

Under the Method of Equal Shares, each resident receives an equal share of the budget, and a project is funded only if its supporters can jointly pay for it from their shares, so that each group of residents gets a fair part of the budget. The team I belong to convinced Polish cities, including Wieliczka and Świecie, to adopt it, and in 2025 I co-founded *Science for Democracy*, a nonprofit that works to bring the rule to Polish municipalities. Direction B builds on this experience.

Toward language. At CNRS LAMSADE, I designed simulation frameworks that model multi-agent decision-making. At the University of Geneva, I study collective decisions in international politics, combining voting records with NLP pipelines that extract structured information from large collections of diplomatic and policy texts.

Simulating democratic deliberation with LLM agents. With collaborators at the University of Toronto and Google DeepMind, we built *VoteSim*, which simulates a full democratic cycle with LLM agents [C28]. Voters, conditioned on the profiles and statements of real participants in the PRISM dataset, rank the parties' policies on twelve divisive issues; seven electoral systems turn the same ballots into seven parliaments; and each parliament negotiates a bill, which the voters then evaluate. Across all seven language models we tested, voters preferred the bills of proportional systems to those of majoritarian ones, and the least satisfied fifth of voters gained the most. Because the voters are simulated, these results describe language-model agents, not real electorates.

What an abstention means in the UN General Assembly. Abstention is the most common way states withhold support in the General Assembly (11.8% of roughly 860,000 votes in 1946–2025, against 5.5% votes against), yet standard measures of state preferences treat every abstention alike. In ongoing work, we classify all 101,070 abstentions by context (cast alone, inside a large crowd of abstainers, or against an organized opposing camp), using only the other states' votes and the abstaining state's voting bloc. Abstaining alone is concentrated among the permanent members of the Security Council, and the mix of contexts tracks international conflict while the overall abstention rate does not. We label the subject of each resolution by embedding its text with a language model and matching it to the UN library's thesaurus.

Research Agenda

Social choice models treat the alternatives as abstract symbols, whereas a city project, a comment in an online consultation, or a UN resolution is a text with content. Two recent results deal with text directly. Fish et al. (2024) prove proportionality guarantees for slates of LLM-generated statements, assuming that the LLM answers its queries accurately. Tessler et al. (2024, in *Science*) found that, in groups of five participants, statements written by an LLM mediator were accepted more often than those of human mediators. In a position paper written with colleagues after a Dagstuhl seminar, we chart how text can enter every stage of collective decision making [P1]. What is still missing are ways to *evaluate* text-based outcomes and *real data* to evaluate them on. My work provided both for ballots. My agenda has three connected research directions, each continuing one of the three lines of work above.

Direction A: A geometry of collective decision problems, from ballots to text.

- **Computing invariant distances at scale.** Which domains admit polynomial-time relaxations of isomorphism-invariant distances that, like the positionwise distance, preserve the geometry? Can sketching and sublinear techniques make them scale to elections with millions of voters?
- **Benchmarks with provable coverage.** How many instances, and which ones, must a benchmark contain to cover the space of instances up to a given precision under a given distance? Which samplers achieve this? The answers will become defaults in `prefsampling` and `mapof`.
- **Maps of deliberation.** When alternatives are texts, two decision problems can be similar in how people voted, in what the alternatives say, or both. Distances that combine the two will make it possible to compare deliberations, human or AI, the way we now compare elections, and to place sessions of the UN General Assembly on a common map. Direction C will use these distances to compare text-based outcomes.

Direction B: Participatory democracy as a living laboratory.

- **Votes over texts.** I will extend Pabulib with project descriptions, residents' comments, and outcomes over several years, turning it into a longitudinal benchmark in which every ballot refers to a text. The UN General Assembly record, with 5,693 resolutions adopted by recorded vote together with their full texts, is a second testbed of this kind. Direction C will use both as real data.
- **Does proportionality carry over to content?** Proportional rules for participatory budgeting have been observed to spread funds evenly across project categories (Nelissen, 2023; Peters and Skowron, 2025). With project texts, we can ask whether they also give each group of residents a proportional share of the *topics* it cares about, and design rules that guarantee it.
- **Does the rule change the process?** We have analyzed how proposers choose project costs under the Method of Equal Shares [C23]. Field studies with partner cities can test how the rule affects real proposers and participants: who proposes, what they propose, and who votes.
- **Tools for cities and residents.** Open tools to choose, simulate, and audit PB rules and to *explain* their outcomes, so that cities can adopt evidence-based designs and residents can see why a project was or was not funded.

I will seek partnerships with Swiss municipalities for the field studies, building on my experience of working with Polish cities through Science for Democracy.

Direction C: Representation when opinions are expressed in text and aggregated by AI.

- **Auditing summaries and consensus statements.** Given opinions written in natural language and an aggregate text, such as a summary or a consensus statement, how should we measure how proportionally the aggregate represents the population? Which measures satisfy natural axioms, and which can be estimated with few LLM queries and provable error bounds? Current methods *generate* such texts, but we have few principled ways to *evaluate* them.
- **An LLM aggregator is a voting rule.** Fix a language model and a prompt that turns a set of opinions into k statements, and we obtain a multi-winner voting rule whose input is text. Its

properties can then be tested like those of any other rule: does the outcome change when the same opinions are presented in a different order (anonymity)? Can a participant steer it by writing a manipulative opinion (manipulability)? Does it cover each group in proportion to its size? I helped establish this kind of large-scale numerical experiment for classical rules [C15].

- **Polarization among LLM agents.** When LLM agents deliberate and vote repeatedly, under which electoral systems and discussion structures do their positions converge, fragment, or polarize? Polarization indices for elections [C10, C27] provide the measures, and VoteSim [C28] provides the simulation environment.

Goal for the first five years. By the end of this period, my group will have produced three things: maps that let researchers compare deliberations the way they now compare elections (A); a public benchmark of votes over texts, built from participatory budgets and the UN General Assembly record (B); and open tools, with known guarantees, for auditing whether AI-written summaries and consensus statements represent the people they summarize (C). These are the same kinds of contributions that my earlier work made for ballots: data, measures, and an experimental method.

Students, funding, and service. Possible first PhD topics are scalable invariant distances (A, theoretical), proportional coverage of topics in participatory budgeting (B, applied and data-centric), and testing LLM aggregators as voting rules (C, empirical and NLP-oriented). I have already co-supervised more than ten students, led a grant as principal investigator (NCN Preludium), and worked within an ERC-funded project. I will apply to the Swiss National Science Foundation early on and for an ERC Starting Grant built around Direction C. I serve on the editorial board of JAIR (since 2026) and review for NeurIPS, AAAI, and IJCAI.

Why EPFL

EPFL does not have a group in computational social choice, but it is strong in algorithms, natural language processing, and secure systems, which my agenda relies on. It connects in particular with the work of three colleagues:

- *Michael Kapralov*'s sublinear algorithms and sketching, for making isomorphism-invariant distances computable on elections with millions of voters (Direction A);
- *Bryan Ford*'s work on proof of personhood and coercion-resistant e-voting, for digital participatory processes that count each resident once and protect the secrecy of the vote (Direction B);
- *Antoine Bosselut*'s NLP group and EPFL's work on open multilingual language models such as Apertus, for auditing whether AI-written summaries represent people who write in different languages, which matters in a country with four national languages (Direction C).

I also want to do this work in Switzerland because of its strong tradition of voting. Swiss citizens decide federal, cantonal, and municipal questions by popular vote several times a year, and in 2023 the city of Aarau used the Method of Equal Shares for its participatory budget. Geneva, where I already work, hosts the UN Human Rights Council, whose votes could extend my work on the General Assembly.

Other references. Fish, Gölz, Parkes, Procaccia, Rusak, Shapira, and Wüthrich. Generative social choice. *Proceedings of EC-2024*. Nelissen. An empirical analysis of participatory budgeting in Amsterdam. arXiv:2310.18033, 2023. Peters and Skowron. The Method of Equal Shares for participatory budgeting. `https://equalshares.net`, 2025. Tessler, Bakker, Jarrett, et al. AI can help humans find common ground in democratic deliberation. *Science* 386(6719), 2024.